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## *ADVANCING GRAPH REPRESENTATION LEARNING FOR SCIENTIFIC FRACTURE PHYSICS GRAPHICAL DATA*

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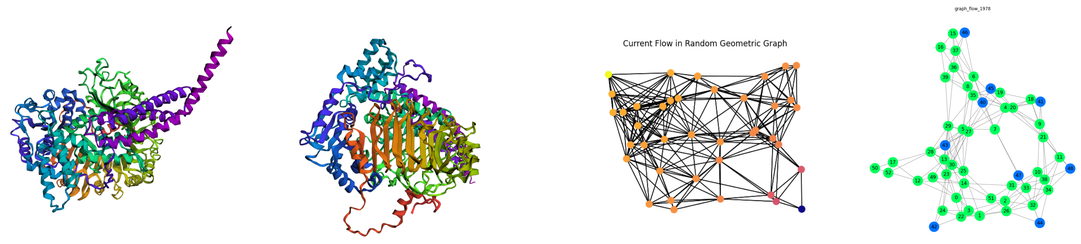
## INTRODUCTION

We intend to extend graph representation learning to what is called a graph transformer on the physical data of porous media. The graph data set from the computation suites Discrete Fracture Network (DFN) is employed to simulate flow and transport in such porous fracture media at scales ranging from millimeters to kilometers. We consider the graphical representation of DFN where nodes represent fractures that is further comprised by certain topological and physical attributes, and edges denote fracture intersections.

**FIGURE 1**: (*Sequentially*) An advanced illustration of graphs: The complex protein structure of *Capsicum Annum*, *Nicotiana tabacum* followed by *Random Geometric Graph (RGG)* and finally the fracture graph of DFNWorks with graph flow index 1978. The protein structure database corresponds to *AlphaFold Protein Structure Database*.




## PROBLEM STATEMENT

The model variations of the transformer Vaswani, 2017 find useful, concrete, and direct applications in computer vision, programming language and to name a few. However, transformers have still not been the de facto standard for DFNWorks at the node level pressure variable prediction. It is still an open question whether Transformer is suitable to model graph representation learning for the aforementioned graphical set.

**FIGURE 2**: Important stages for the proposed Graph-In-Fracture Transformer powered by the Kolmogorov-Arnold Superposition Theorem Layer. Proposed model is designed to work effectively and consistently on a large number of graphical fracture data sets and sits in the realm of small language model computed on LANL Apple M2 MAX 32 GB.

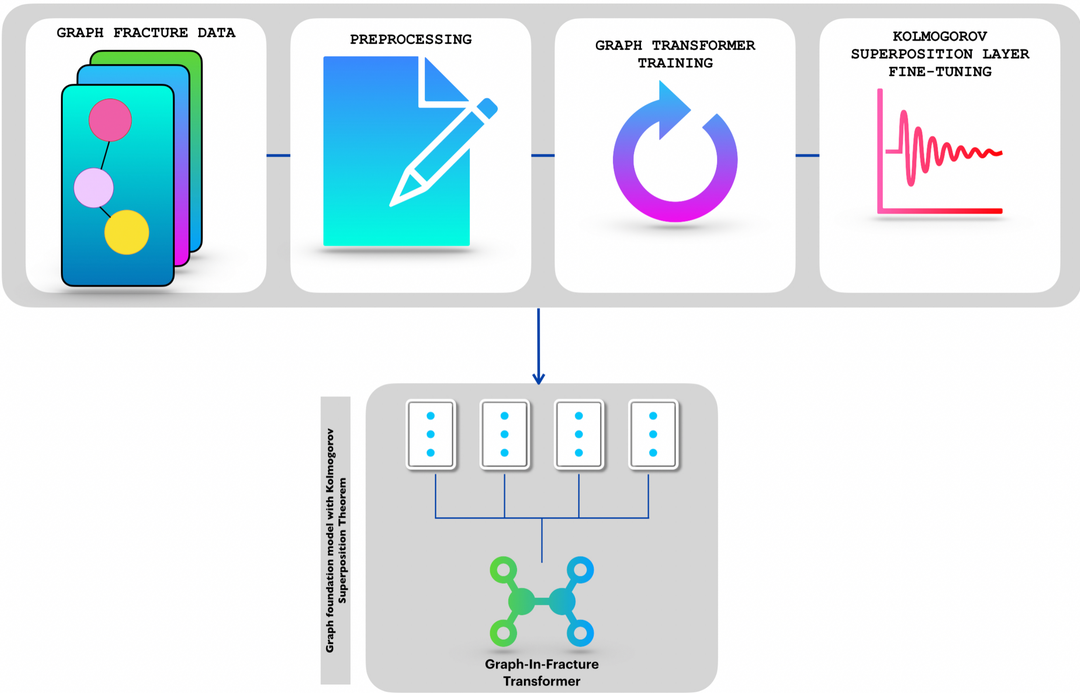



**TABLE 1**

Merit of DFNWorks data set in terms of total number of nodes and edges. Here *all points* in second column imply the total node features times the total nodes summand all over the graphical data set.

| Graphs | All points | All Nodes | Avg Node | All Edges | Avg Edge |
|---|---|---|---|---|---|
| 731 | $\approx 3e7$ | 36,580 | 50.18 | 107,091 | 146.9 |

## METHOD

At every node of the graphical structure of DFN, we have both geometrical variables and hydrological variables. We then constitute the node feature matrix based on these variables to propagate the aforementioned graph neural networks (GNNs).

We incorporated three graph representation learning models namely GCN Kipf and Welling, 2017, GAT, Veličković et al., 2018 and GT, Dwivedi and Bresson, 2021. We incorporated Kolmogorov-Arnold Superposition Theorem Layer (KASTL Kolmogorov, 1957) in GT to perform the linear approximation of the predictions made by the GT to enhance the overall performance of the model.

## GRAPH TRANSFORMER

Let $G$ be a graph with node feature matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$.

$$\textsc{Attention}\left(\mathbf{X}^{(\ell)}\right) := \mathrm{softmax}\left(d_K^{-1/2} Q K^\top\right) V,$$

$d_K$ denotes the feature dimension of the linear transformations $Q$ and $K$, with $\mathbf{X}^{(0)} := \mathbf{X}$. Then, MHA $(\mathbf{X}^{(\ell)})$ concatenates multiple attention heads, followed by an output projection to the feature space of $\mathbf{X}^{(\ell)}$. Finally, the transformer layer updates features $\mathbf{X}^{(\ell)}$ via

$$\mathbf{X}^{(\ell+1)} := \mathrm{FFN}\left(\mathrm{MHA}(\mathbf{X}^{(\ell)}) + \mathbf{X}^{(\ell)}\right).$$

**FIGURE 3**: The novel KASTL used for supervised tuning result for GT to elevate the overall performance of the proposed GFM exhibiting *provable prediction guarantee phenomena (PPGP)*.

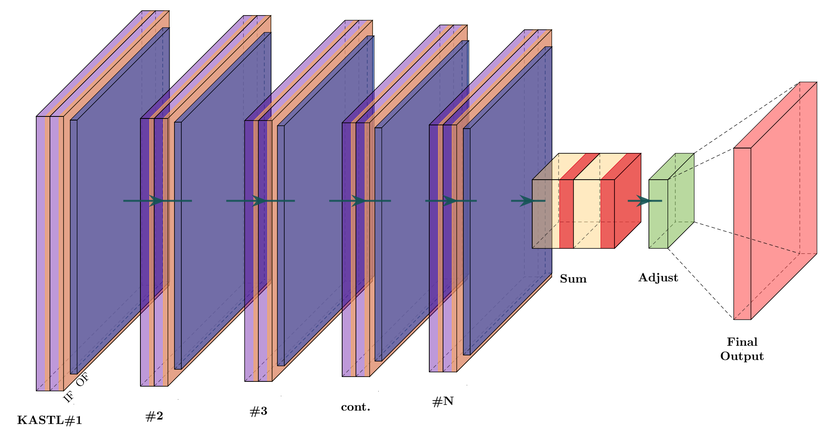
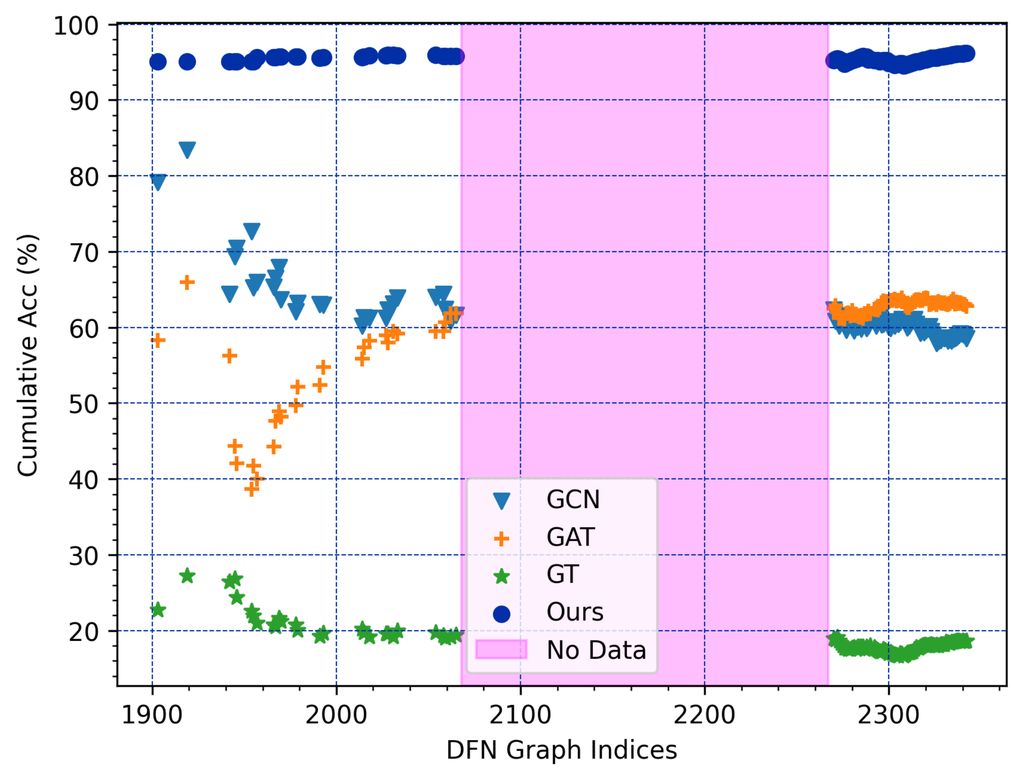



**TABLE 2**

Comparative analysis of all four graph based models during the course of training with respect to the available DFNWorks graphical indices. The accuracy uncertainty in fourth column is the 95% confidence interval which is the industry standard metric method. (Values closer to 100%, corresponds to better model; also refer Figure 4.)

| Model | Min Acc | Max Acc | Acc Uncert. | Final Acc |
|---|---|---|---|---|
| *Ours* | 94.54 | 97.35 | $96.50 \pm 0.62$ | 97.185 |
| GT | 16.68 | 27.27 | $17.97 \pm 0.98$ | 17.16 |
| GCN | 57.89 | 83.41 | $61.45 \pm 3.83$ | 58.53 |
| GAT | 38.69 | 66.00 | $60.20 \pm 5.81$ | 62.79 |

**FIGURE 4**: Cumulative accuracies of all considered four models during the training on finite sample of DFNWorks data is plotted on $y-$axis. We see that the accuracies registered by GCN and GAT are identical as it is allowed to run on every DFNWorks graphical indices. We also infer that GT is able to perform well as training time progresses with the incorporation of KASTL. (*Higher $y-$values implies the better corresponding model is.*)




## FUTURE DIRECTIONS

- Improve and investigate the model accuracies of GT without PPGP.
- Fractured data heterogeneity is definitely a topic of ongoing research.
- Investigation for the robustness of proposed FM.

## ACKNOWLEDGMENT

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